Included Variable Bias in Blair, Di Salvatore & Smidt (2023): An Empirical Application

2026-02-27

1 Introduction

Blair, Di Salvatore, and Smidt (2023, APSR) estimate the effect of UN peacekeeping on democratization in conflict-affected countries using a country fixed effects panel regression. The main specification is:

$$y_{it} = \beta\, D_{i,t-2} + \gamma' \mathbf{Z}_{i,t-3} + \delta_i + \varepsilon_{it}$$

where y_{it} is electoral democracy (V-Dem polyarchy index), $D_{i,t-2}$ is a peacekeeping measure lagged 2 periods (e.g., democracy mandate dummy, personnel counts), $\mathbf{Z}_{i,t-3}$ is a vector of six controls lagged 3 periods (population, foreign aid, GDP per capita, refugees/IDPs, literacy, fuel exports), and δ_i are country fixed effects. Controls are imputed with within-country means before lagging.

A key question is whether any of these controls are **colliders** — variables caused by both the treatment (D) and the outcome (y). If so, conditioning on them introduces **Included Variable Bias** (IVB). The IVB for each candidate collider z is given by:

$$\text{IVB}(\hat{\beta}) = \hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \cdot \hat{\pi}$$

where $\hat{\theta}^*$ is the coefficient of z in the "long" model (with z) and $\hat{\pi}$ is the coefficient of the treatment in the auxiliary regression of z on the treatment and other controls.

An important design feature of Blair et al. is that controls are lagged 3 periods while the treatment is lagged only 2 periods. This temporal ordering means controls are measured *before* the treatment, providing substantial protection against collider bias even for controls where the outcome-to-control causal channel is theoretically plausible.

We apply the IVB decomposition to Blair et al.'s Table 2 (democracy mandate, full sample), treating each control as a candidate collider.

2 Data

```r
df_raw <- read_dta("candidate_papers/peacekeeping/data_replication.dta")

# --- Replicate Stata data processing steps ---

# Drop Eritrea before 1993 and South Sudan before 2011
df <- df_raw %>%
  filter(!(gwnoloc == 531 & year < 1993)) %>%
  filter(!(gwnoloc == 626 & year < 2011))

# Sort by country-year
df <- df %>% arrange(gwnoloc, year)
```

```r
# --- Generate lagged dependent variables ---
df <- df %>%
  group_by(gwnoloc) %>%
  mutate(
    v2x_polyarchy_1l = lag(v2x_polyarchy, 1),
    v2x_polyarchy_2l = lag(v2x_polyarchy, 2),
    v2x_polyarchy_3l = lag(v2x_polyarchy, 3)
  ) %>%
  ungroup()

# --- Divide personnel numbers by 1,000 ---
df <- df %>%
  mutate(
    itotal_compound_K = itotal_compound / 1000,
    iactual_civilian_total_K = iactual_civilian_total / 1000
  )

# --- Generate lagged treatment variables (2-period lag) ---
df <- df %>%
  group_by(gwnoloc) %>%
  mutate(
    ipema_any_demo_assist_dum_2l = lag(ipema_any_demo_assist_dum, 2),
    itotal_compound_K_2l = lag(itotal_compound_K, 2),
    iactual_civilian_total_K_2l = lag(iactual_civilian_total_K, 2),
    iany_demo_all_max_dum_2l = lag(iany_demo_all_max_dum, 2)
  ) %>%
  ungroup()

# --- Impute missing controls with within-country means ---
ctrl_vars <- c("wdi_pop", "wdi_oda", "wdi_gdppc", "unhcr_ref_idp",
               "wdi_literacy", "wdi_fuel")

for (v in ctrl_vars) {
  new_v <- paste0("i", v)
  df[[new_v]] <- df[[v]]
  country_means <- df %>%
    group_by(gwnoloc) %>%
    summarise(cmean = mean(.data[[v]], na.rm = TRUE), .groups = "drop")
  df <- df %>%
    left_join(country_means, by = "gwnoloc", suffix = c("", "_cm"))
  df[[new_v]] <- ifelse(is.na(df[[new_v]]), df$cmean, df[[new_v]])
  df$cmean <- NULL
}

# --- Generate lagged control variables (3-period lag, imputed) ---
df <- df %>%
  group_by(gwnoloc) %>%
  mutate(
    iwdi_pop_3l = lag(iwdi_pop, 3),
    iwdi_oda_3l = lag(iwdi_oda, 3),
    iwdi_gdppc_3l = lag(iwdi_gdppc, 3),
    iunhcr_ref_idp_3l = lag(iunhcr_ref_idp, 3),
    iwdi_literacy_3l = lag(iwdi_literacy, 3),
```

```r
    iwdi_fuel_3l = lag(iwdi_fuel, 3)
  ) %>%
  ungroup()

# --- Drop observations before 1991 ---
df <- df %>% filter(year >= 1991)
```

The dataset contains 1154 country-year observations for 42 countries over 1991–2018.

3 Replication of Table 2 (Democracy Mandate)

```r
y_var <- "v2x_polyarchy"
d_var <- "ipema_any_demo_assist_dum_2l"
ctrl_vars_lagged <- c("iwdi_pop_3l", "iwdi_oda_3l", "iwdi_gdppc_3l",
                      "iunhcr_ref_idp_3l", "iwdi_literacy_3l", "iwdi_fuel_3l")
fe_var <- "gwnoloc"

fml_t2 <- as.formula(paste0(y_var, " ~ ", d_var, " + ",
                            paste(ctrl_vars_lagged, collapse = " + "),
                            " | ", fe_var))
m_t2 <- feols(fml_t2, data = df, vcov = "iid")
```

```r
t2_coefs <- data.frame(
  Term = names(coef(m_t2)),
  Estimate = as.numeric(round(coef(m_t2), 4)),
  SE = as.numeric(round(sqrt(diag(vcov(m_t2))), 4))
)
kable(t2_coefs, digits = 4,
      caption = "Table 2 Column 1: Electoral Democracy and UN Democracy Mandates (Full Sample)")
```

Table 1: Table 2 Column 1: Electoral Democracy and UN Democracy Mandates (Full Sample)

Term	Estimate	SE
ipema_any_demo_assist_dum_2l	0.1184	0.0130
iwdi_pop_3l	0.0000	0.0000
iwdi_oda_3l	0.0000	0.0000
iwdi_gdppc_3l	0.0000	0.0000
iunhcr_ref_idp_3l	0.0000	0.0000
iwdi_literacy_3l	0.0002	0.0013
iwdi_fuel_3l	0.0001	0.0004

The democracy mandate coefficient is $\hat{\beta} = 0.118$ (SE = 0.013, N = 832). Countries with UN democracy mandates have higher subsequent electoral democracy scores, consistent with the original results.

4 IVB Decomposition

4.1 Setup

We define:

- **Treatment** (d): `ipema_any_demo_assist_dum_2l` (democracy mandate, 2-year lag)

- **Outcome** (y): `v2x_polyarchy` (V-Dem electoral democracy index)
- **Candidate colliders** (z): each of the six controls, one at a time
- **Fixed effects**: country (`gwnoloc`)

For each candidate collider z, the "short" model excludes z (keeping the other five controls as legitimate controls w) and the "long" model includes it. The IVB identity $\hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \cdot \hat{\pi}$ is verified numerically.

4.2 Table 2: Democracy Mandate (Full Sample)

```
ctrl_labels <- c(
  "iwdi_pop_3l" = "Population (3yr lag)",
  "iwdi_oda_3l" = "Foreign Aid (3yr lag)",
  "iwdi_gdppc_3l" = "GDP per capita (3yr lag)",
  "iunhcr_ref_idp_3l" = "Refugees & IDPs (3yr lag)",
  "iwdi_literacy_3l" = "Literacy (3yr lag)",
  "iwdi_fuel_3l" = "Fuel Exports (3yr lag)"
)

ivb_t2 <- lapply(ctrl_vars_lagged, function(z_var) {
  w_vars <- setdiff(ctrl_vars_lagged, z_var)
  compute_ivb_multi(
    data = df, y = y_var, d_vars = d_var,
    z = z_var, w = w_vars,
    fe = fe_var, vcov = "iid"
  )
})
names(ivb_t2) <- ctrl_vars_lagged

tab_t2 <- do.call(rbind, lapply(names(ivb_t2), function(z) {
  r <- ivb_t2[[z]]$results
  data.frame(
    Collider = ctrl_labels[z],
    beta_short = r$beta_short,
    beta_long = r$beta_long,
    theta = r$theta,
    pi = r$pi,
    IVB = r$ivb_formula,
    check = r$diff_check,
    stringsAsFactors = FALSE
  )
}))
```

```
tab_t2_display <- data.frame(
  Collider   = tab_t2$Collider,
  beta_short = sprintf("%.4f", tab_t2$beta_short),
  beta_long  = sprintf("%.4f", tab_t2$beta_long),
  theta      = sprintf("%.2e", tab_t2$theta),
  pi         = sprintf("%.2e", tab_t2$pi),
  IVB        = sprintf("%.4f", tab_t2$IVB),
  check      = sprintf("%.2e", tab_t2$check),
  stringsAsFactors = FALSE
)

kable(tab_t2_display, row.names = FALSE, align = "lrrrrrr",
```

```
      caption = "IVB decomposition for each candidate collider (Table 2, Democracy Mandate)",
      col.names = c("Candidate collider $z$", "$\\hat\\beta_{\\text{short}}$",
                    "$\\hat\\beta_{\\text{long}}$", "$\\hat\\theta^*$",
                    "$\\hat\\pi$", "IVB $= -\\hat\\theta^* \\hat\\pi$",
                    "Check"))
```

Table 2: IVB decomposition for each candidate collider (Table 2, Democracy Mandate)

Candidate collider z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB $= -\hat{\theta}^*\hat{\pi}$	Check
Population (3yr lag)	0.1189	0.1184	3.84e-09	1.41e+05	-0.0005	-1.83e-17
Foreign Aid (3yr lag)	0.1155	0.1184	-7.55e-12	3.83e+08	0.0029	5.64e-18
GDP per capita (3yr lag)	0.1128	0.1184	2.38e-05	-2.32e+02	0.0055	-1.73e-17
Refugees & IDPs (3yr lag)	0.1134	0.1184	-2.76e-08	1.80e+05	0.0050	-1.99e-17
Literacy (3yr lag)	0.1183	0.1184	2.16e-04	-1.83e-01	0.0000	-2.66e-17
Fuel Exports (3yr lag)	0.1194	0.1184	1.44e-04	6.91e+00	-0.0010	-3.04e-17

The identity holds exactly (check ≈ 0) in all cases. GDP per capita produces the largest IVB (0.0055), followed by refugees/IDPs (0.005) and foreign aid (0.0029).

4.3 IVB as Proportion of Total

```
total_ivb <- sum(tab_t2$IVB)

tab_pct <- tab_t2 %>%
  select(Collider, IVB) %>%
  arrange(desc(abs(IVB)))

tab_pct$IVB_display <- sprintf("%.4f", tab_pct$IVB)
tab_pct$Pct_of_total <- sprintf("%.1f%%",  100 * tab_pct$IVB / total_ivb)

tab_pct_display <- tab_pct %>% select(Collider, IVB_display, Pct_of_total)

kable(tab_pct_display, row.names = FALSE,
      caption = "IVB magnitudes sorted by absolute value",
      col.names = c("Control", "IVB", "\\% of total IVB"))
```

Table 3: IVB magnitudes sorted by absolute value

Control	IVB	% of total IVB
GDP per capita (3yr lag)	0.0055	46.4%
Refugees & IDPs (3yr lag)	0.0050	41.9%
Foreign Aid (3yr lag)	0.0029	24.3%
Fuel Exports (3yr lag)	-0.0010	-8.4%
Population (3yr lag)	-0.0005	-4.5%
Literacy (3yr lag)	0.0000	0.3%

The total IVB across all six controls is 0.0119. GDP per capita and refugees/IDPs together account for the bulk of the total bias.

5 Literature Support for Collider Status

The IVB formula diagnoses **how much** each control shifts the treatment estimate, but **whether** a control is a collider depends on substantive knowledge: does the outcome y_{it} (electoral democracy) cause the control z_{it}?

5.1 Democracy → GDP per capita

There is strong evidence that democracy causes economic growth:

- **Acemoglu, Naidu, Restrepo & Robinson (2019, JPE):** Democratization increases GDP per capita by approximately 20% over 25 years. Uses dynamic panel (ADL), system GMM, and IV (regional waves of democratization). This is the current consensus result.
- **Papaioannou & Siourounis (2008, EJ):** Democratization associated with approximately 1 percentage point per year of additional growth.
- **Doucouliagos & Ulubasoglu (2008, AJPS):** Meta-analysis of 84 studies finds robust indirect effects of democracy on growth via human capital, inflation, and economic freedom.

Implication: GDP per capita is a plausible collider. However, the 3-year lag structure means GDP at $t-3$ cannot be caused by democracy at t. The concern is limited to persistent within-country democracy trajectories creating weak feedback effects.

5.2 Democracy → Refugees and IDPs

There is moderate evidence linking democracy to displacement:

- **Davenport (2007, Cambridge UP):** Democracy reduces state repression (the "domestic democratic peace"), which is a primary driver of forced displacement.
- **Iqbal & Zorn (2006, JCR):** Higher levels of democracy reduce refugee outflows.
- **Moore & Shellman (2004, JCR):** Political repression and armed conflict are key push factors for forced migration.

Implication: Refugees/IDPs is a plausible collider in post-conflict settings, where democratic transitions may reduce displacement. However, the relationship is complex during unstable transitions, and the 3-year lag provides protection.

5.3 Democracy → Foreign Aid

There is strong evidence that donors allocate more aid to democracies:

- **Alesina & Dollar (2000, JEG):** Countries that democratize receive more aid.

- **Bermeo (2016, IO):** Aid from democratic donors rewards democratization; heterogeneity across donor types.
- **Savun & Tirone (2011, JCR):** Foreign aid promotes democratization particularly in post-conflict settings.

Implication: Foreign aid is a plausible collider, especially in the peacekeeping context where democratic transitions trigger donor engagement. The 3-year lag attenuates but does not eliminate the concern for persistent democratizers.

5.4 Democracy → Fuel Exports

No evidence. Fuel exports are determined by geological endowments and global commodity prices, not by political regime type. The resource curse literature (Ross 2001, 2012) examines the reverse direction (oil → politics).

5.5 Democracy → Literacy

Weak/long-run evidence. Democracies invest more in education (Stasavage 2005), but literacy is a slow-moving stock variable that changes over decades, not years. Within the time horizons of this panel, the effect is negligible.

5.6 Democracy → Population

No evidence. Population growth is a demographic variable driven by fertility, mortality, and net migration. These respond to political institutions only over very long time horizons (Besley & Kudamatsu 2006).

6 Summary

```r
summary_all <- data.frame(
  Control = tab_t2$Collider,
  `Y causes Z` = c("No (demographic)", "Strong (Bermeo 2016)",
                   "Strong (Acemoglu et al. 2019)", "Moderate (Davenport 2007)",
                   "Weak/long-run", "No (geological)"),
  `Key reference` = c("---", "Alesina & Dollar 2000",
                      "Acemoglu et al. 2019", "Iqbal & Zorn 2006",
                      "Stasavage 2005", "Ross 2001"),
  `IVB` = round(tab_t2$IVB, 4),
  `Lag protects` = c("Yes (fully)", "Substantially", "Substantially",
                     "Substantially", "Yes (fully)", "Yes (fully)"),
  check.names = FALSE, stringsAsFactors = FALSE
)
kable(summary_all, caption = "Summary: Literature evidence, IVB magnitudes, and lag protection")
```

Table 4: Summary: Literature evidence, IVB magnitudes, and lag protection

Control	Y causes Z	Key reference	IVB	Lag protects
Population (3yr lag)	No (demographic)	—	-0.0005	Yes (fully)
Foreign Aid (3yr lag)	Strong (Bermeo 2016)	Alesina & Dollar 2000	0.0029	Substantially
GDP per capita (3yr lag)	Strong (Acemoglu et al. 2019)	Acemoglu et al. 2019	0.0055	Substantially

Control	Y causes Z	Key reference	IVB	Lag protects
Refugees & IDPs (3yr lag)	Moderate (Davenport 2007)	Iqbal & Zorn 2006	0.0050	Substantially
Literacy (3yr lag)	Weak/long-run	Stasavage 2005	0.0000	Yes (fully)
Fuel Exports (3yr lag)	No (geological)	Ross 2001	-0.0010	Yes (fully)

Three of the six controls — GDP per capita, refugees/IDPs, and foreign aid — have documented causal links from the outcome (democracy). These three controls produce the largest IVBs. However, the 3-year lag structure substantially mitigates collider concerns for all controls:

1. **GDP per capita (IVB = 0.0055):** Largest IVB component. Democracy does cause growth (Acemoglu et al. 2019), but the 3-year lag means GDP at $t-3$ precedes both the treatment ($t-2$) and the outcome (t), limiting the collider channel.
2. **Refugees & IDPs (IVB = 0.005):** Second-largest IVB. Democracy may reduce displacement (Davenport 2007), but the temporal ordering provides protection.
3. **Foreign aid (IVB = 0.0029):** Donors reward democratizers (Bermeo 2016), but the 3-year lag attenuates the concern.
4. **Fuel exports, literacy, population:** These have near-zero IVBs and no plausible collider channels.

Blair et al.'s design choice to lag controls an additional period beyond the treatment (3 vs. 2 years) is a deliberate and effective strategy to mitigate collider bias. The IVB decomposition reveals the sensitivity of the treatment coefficient to each control but, given the lag structure, the estimated IVBs likely overstate actual collider bias.

7 References

- Acemoglu, D., Naidu, S., Restrepo, P., & Robinson, J. A. (2019). Democracy Does Cause Growth. *Journal of Political Economy*, 127(1), 47–100.
- Alesina, A. & Dollar, D. (2000). Who Gives Foreign Aid to Whom and Why? *Journal of Economic Growth*, 5(1), 33–63.
- Bermeo, S. B. (2016). Aid Is Not Oil: Donor Utility, Heterogeneous Aid, and the Aid-Democratization Relationship. *International Organization*, 70(1), 1–32.
- Besley, T. & Kudamatsu, M. (2006). Health and Democracy. *American Economic Review*, 96(2), 313–318.
- Blair, R. A., Di Salvatore, J., & Smidt, H. M. (2023). UN Peacekeeping and Democratization in Conflict-Affected Countries. *American Political Science Review*, 117(4), 1347–1367.
- Davenport, C. (2007). *State Repression and the Domestic Democratic Peace*. Cambridge University Press.
- Doucouliagos, H. & Ulubasoglu, M. A. (2008). Democracy and Economic Growth: A Meta-Analysis. *American Journal of Political Science*, 52(1), 61–83.
- Iqbal, Z. & Zorn, C. (2006). Swords into Ploughshares: United Nations Peacebuilding and Refugee Return. *Journal of Conflict Resolution*, 50(2), 227–249.
- Moore, W. H. & Shellman, S. M. (2004). Fear of Persecution: Forced Migration, 1952–1995. *Journal of Conflict Resolution*, 48(5), 723–745.
- Papaioannou, E. & Siourounis, G. (2008). Democratisation and Growth. *The Economic Journal*, 118(532), 1520–1551.
- Ross, M. L. (2001). Does Oil Hinder Democracy? *World Politics*, 53(3), 325–361.
- Ross, M. L. (2012). *The Oil Curse: How Petroleum Wealth Shapes the Development of Nations*. Princeton University Press.
- Savun, B. & Tirone, D. C. (2011). Foreign Aid, Democratization, and Civil Conflict. *American Journal of Political Science*, 55(2), 233–246.

- Stasavage, D. (2005). Democracy and Education Spending in Africa. *American Journal of Political Science*, 49(2), 343–358.